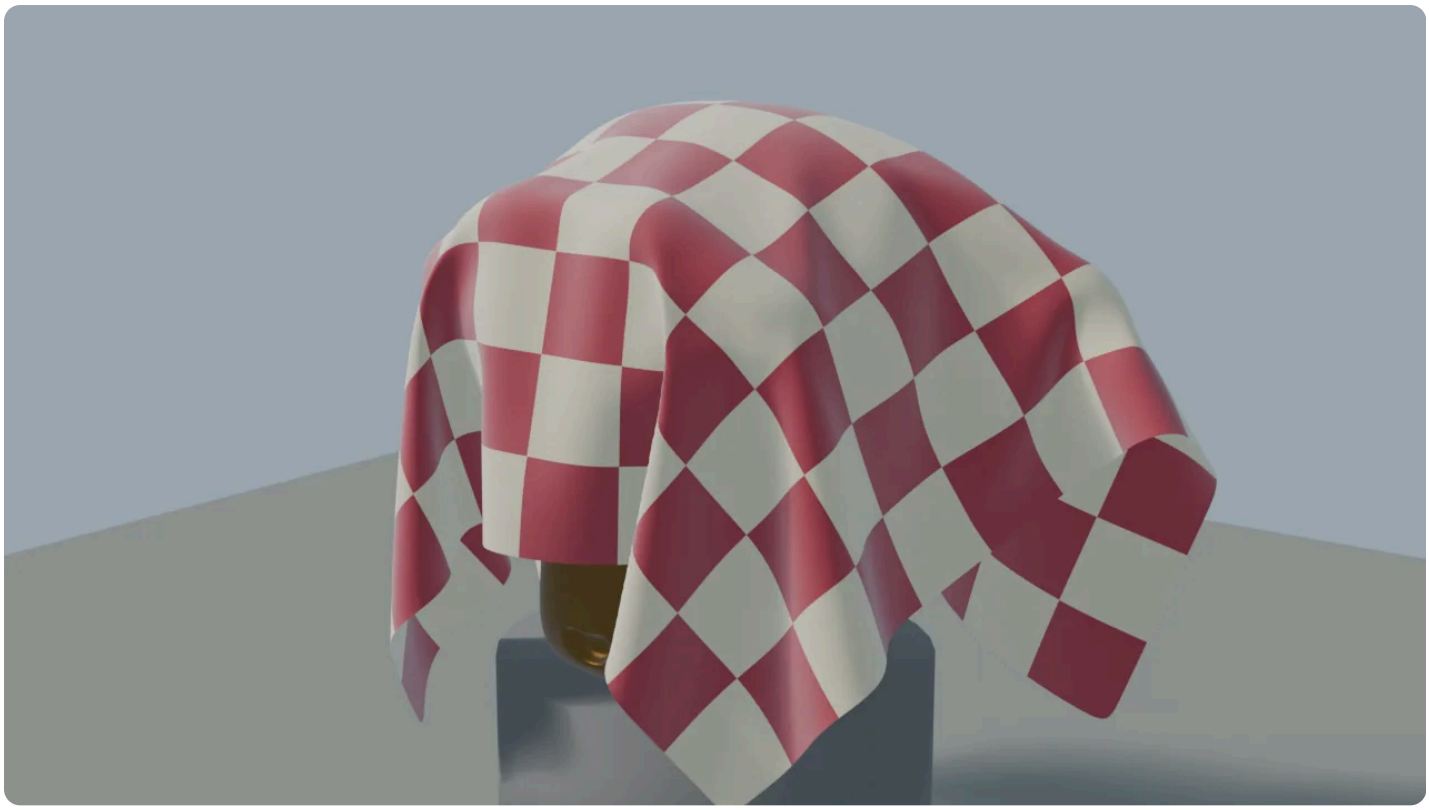


Physics Lab 3

Draped Cloth

Drop a sheet onto Suzanne and make it fall, fold and hang like real fabric in Blender.

Rendered in Blender by following the steps in this lab.

Today

We are learning

how a flat sheet becomes cloth: why it needs vertices to bend, what it needs to land on, which numbers make it silk or felt, and how to make the result look like fabric instead of paper.

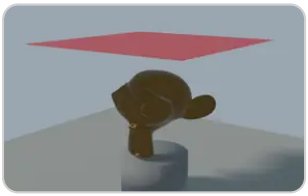
> You need Blender 4.2 or newer and a mouse with a scroll wheel (laptop trackpad? see this fix). This lab takes about two lessons. Save your file as **PhysicsLab3_DrapedCloth_YourName.blend**.

You’ve got it when you can

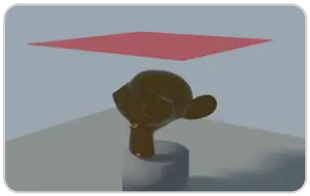
- subdivide a plane and explain why cloth needs vertices
- add Cloth and Collision and drop a sheet onto a statue
- tune collisions, self collisions and bake the result
- say what Bending, Vertex Mass and Quality Steps change
- smooth a simulated mesh with Shade Smooth and Subdivision, in the right order
- give it a fabric material, light it, and render a still and a movie

The plan

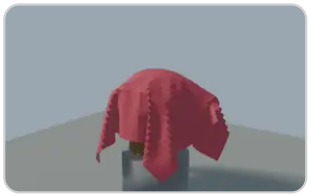
Four parts. Each one ends with a rendered picture of what your scene should look like.



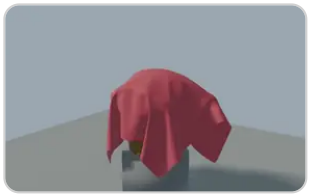
Part 1
A statue on a drum
Steps 1–3



Part 2
A sheet that is cloth
Steps 4–9



Part 3
Smooth it
Steps 10–11

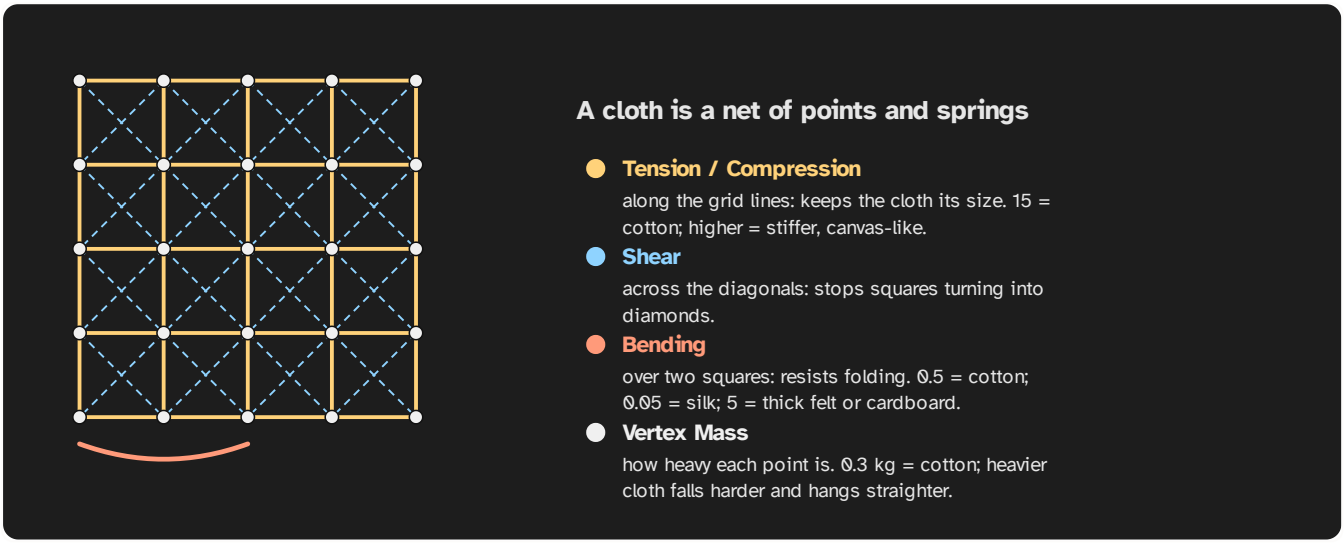


Part 4
Fabric, light, render
Steps 12–15

A net of springs

Blender’s cloth is not a sheet; it is a net. Every vertex is a tiny weight, every edge a tiny spring, and gravity pulls the weights down while the springs pull them back. Where the net meets an object with **Collision**, it stops. That is the whole trick, and it explains everything in this lab: more vertices means more places to bend, stiffer springs mean fewer folds, heavier weights mean a straighter hang.

The simulation only ever moves the real vertices. Smooth shading and a Subdivision Surface modifier are display tricks applied afterwards, so the cloth can be coarse (fast) and still look fine (smooth).



Blender’s cloth is a net of tiny springs. Quality Steps is how many times per frame it checks them: more is calmer and slower.

– Word bank — open this if a word in the steps is new

Subdivide

Cut every face into smaller faces (Edit Mode). Number of Cuts 40 turns one square into 41 × 41.

Cloth

A physics setting that turns a mesh into a net of weights and springs that falls and drapes.

Collision

A physics setting on an object that cloth must not pass through. The object itself does not move.

Quality Steps

How many times per frame the springs are checked. More = calmer, slower.

Vertex Mass

How heavy each point is. 0.3 kg = cotton.

Stiffness: Tension / Shear / Bending

How hard the cloth resists stretching, skewing and folding. Bending is the one that changes the look most.

Self Collisions

Stops the cloth passing through its own folds.

Cache / Bake

Blender's memory of simulated frames; Bake locks it. Delete Bake before changing numbers.

Shade Smooth

Blend the shading across faces so facets vanish. Display only.

Subdivision Surface

A modifier that splits every face into four and rounds the result. Levels 2 = 16 times the faces.

Modifier stack

The list in the wrench tab, applied top to bottom. Order matters.

Checker Texture

A node that makes a two-colour check pattern; Scale is squares across.

Sheen

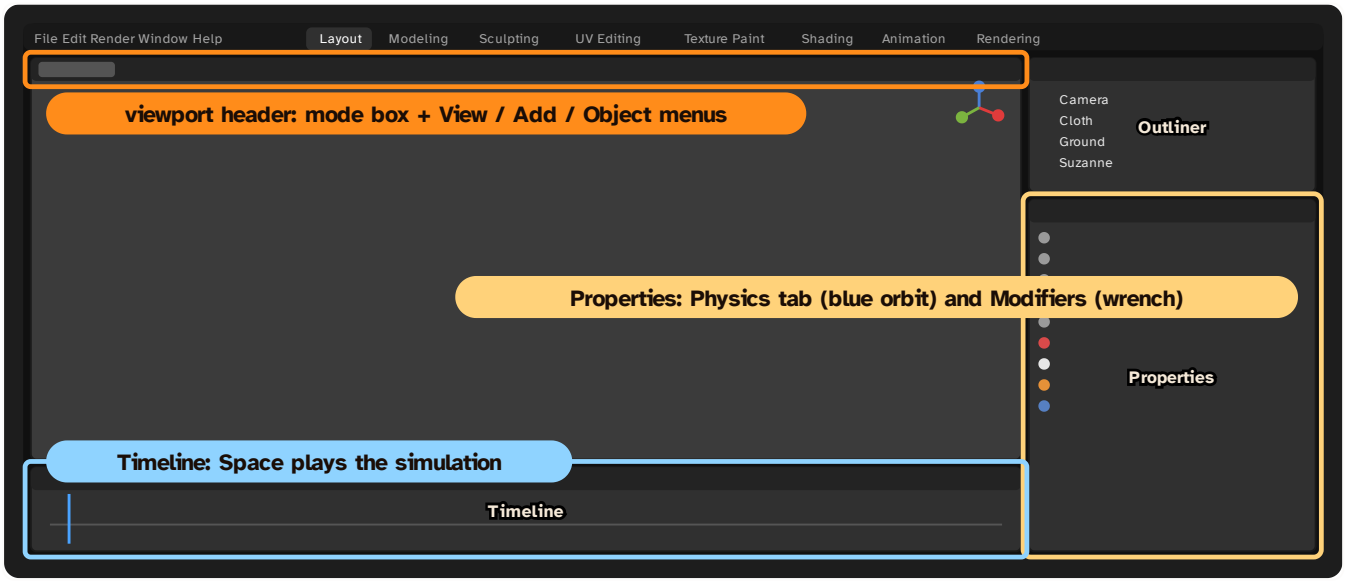
The fuzzy edge glow of fabric, a slider in the Principled BSDF.

Part 1

A statue on a drum

Tools: **Add** › Mesh, the sidebar, **Shade Smooth**, the **Subdivision Surface** modifier.

Something has to be under the sheet. Suzanne, Blender's monkey, stands on a drum on a floor. All three will become collision objects, so the cloth has something to land on.



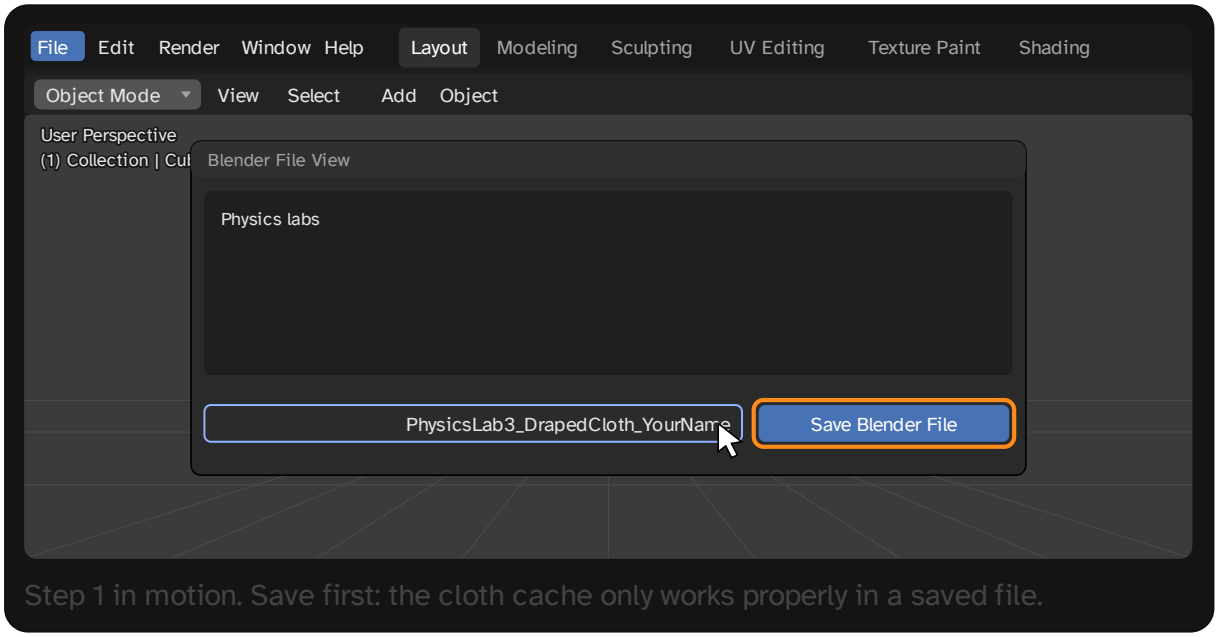
The Layout workspace. Hover the viewport and press N for the sidebar (exact numbers live there).

How the steps work

Each step has a **Where** line (which part of the Blender window to look at), a **Watch** animation of the mouse doing the step (pause it, drag the slider, or use the arrow buttons to go one move at a time), and a **Do this** list of the same moves as a, b, c. Do one move, look at the animation, do the next. The drawings are simplified, not screenshots: the words and positions match Blender, the exact colours and spacing may not. Suzanne in particular is a cartoon of herself.

1 Start fresh and save

Layout workspace › the **File** menu at the very top-left.



Step 1 in motion. Save first: the cloth cache only works properly in a saved file.

Do this:

a **File** › **New** › **General**. You get a cube, a camera and a light.

b Click the cube once (orange outline), press X and click **Delete**. Keep the camera and the light.

c **File** › **Save As**. In your Physics labs folder, type **PhysicsLab3_DrapedCloth_YourName** and click **Save Blender File**.

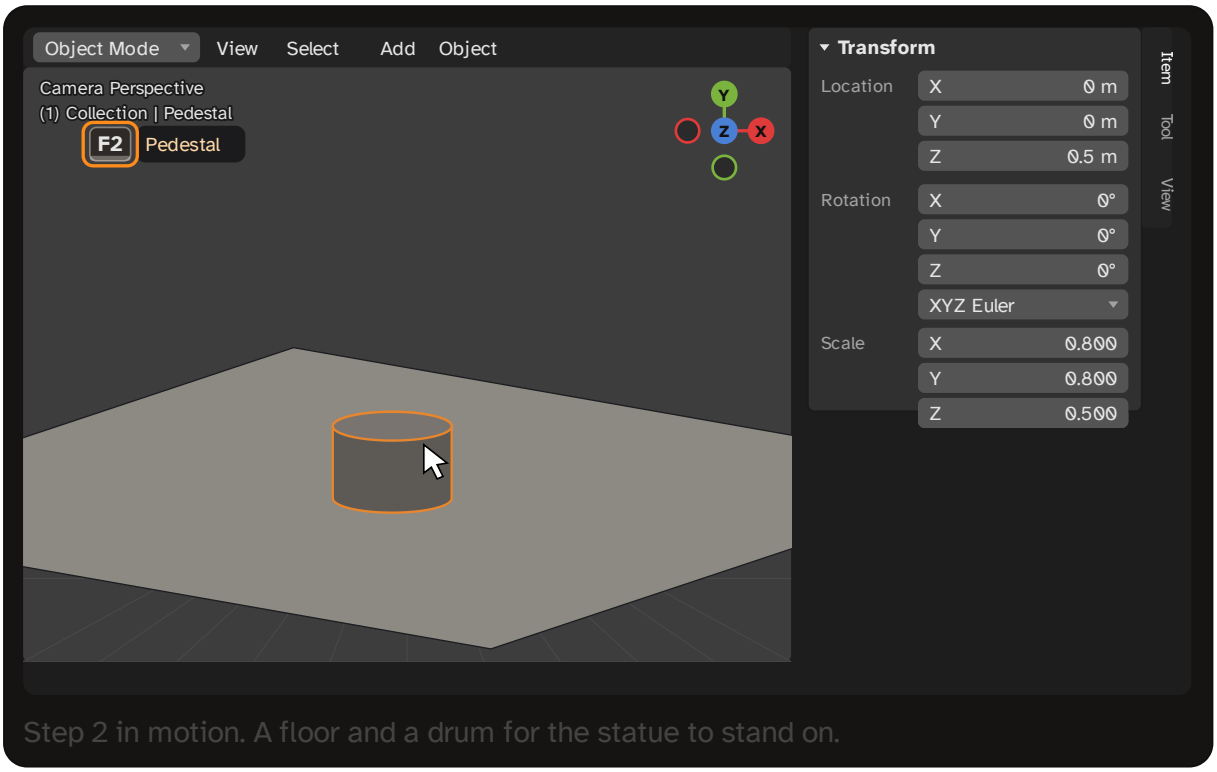
> **You should now see:** the window title shows your file name, and the Outliner (top right) lists only Camera and Light.

Why save first?

> Cloth keeps its simulated frames in a cache next to the file. In an unsaved file some cache options are greyed out and the bake cannot be kept. Save first and everything works.

2 Floor and drum

The **Add** menu, the sidebar (**N**), the right-click menu.



Step 2 in motion. A floor and a drum for the statue to stand on.

Do this:

a **Add** › **Mesh** › **Plane**. S, 6, Enter: a 12 m floor. F2, type **Ground**, Enter.

b **Add** › **Mesh** › **Cylinder**. Sidebar: **Location** Z **0.5**. **Scale** X **0.8**, Y **0.8**, Z **0.5**. A drum 1 m tall and 1.6 m across, standing on the floor.

c Right-click the drum › **Shade Smooth**. F2, type **Pedestal**, Enter. Material tab › New › Hex **6E6A66** (optional, for looks).

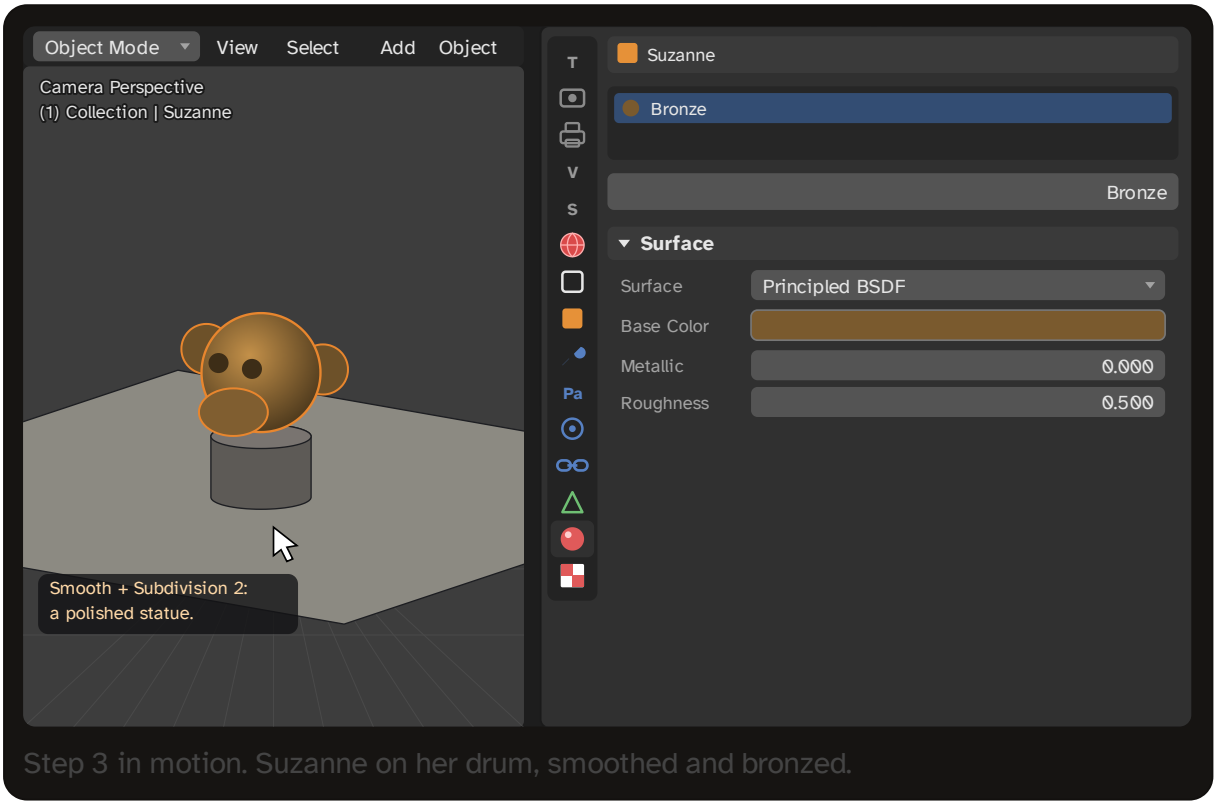
> **You should now see:** a grey floor and a smooth drum in the middle of it.

Why Scale Z 0.5 and Location Z 0.5?

> A new cylinder is 2 m tall, centred on its middle. Scale Z 0.5 makes it 1 m tall; its middle is then 0.5 m up, so Location Z 0.5 stands it exactly on the floor.

3 Suzanne, smoothed and bronzed

Add › Mesh › Monkey, the sidebar, the **Modifiers** (wrench) tab, the **Material** tab.



Step 3 in motion. Suzanne on her drum, smoothed and bronzed.

Do this:

a **Add › Mesh › Monkey**. Sidebar **Location** Z: **2**. She sits on the drum (her chin is 2 cm above it).

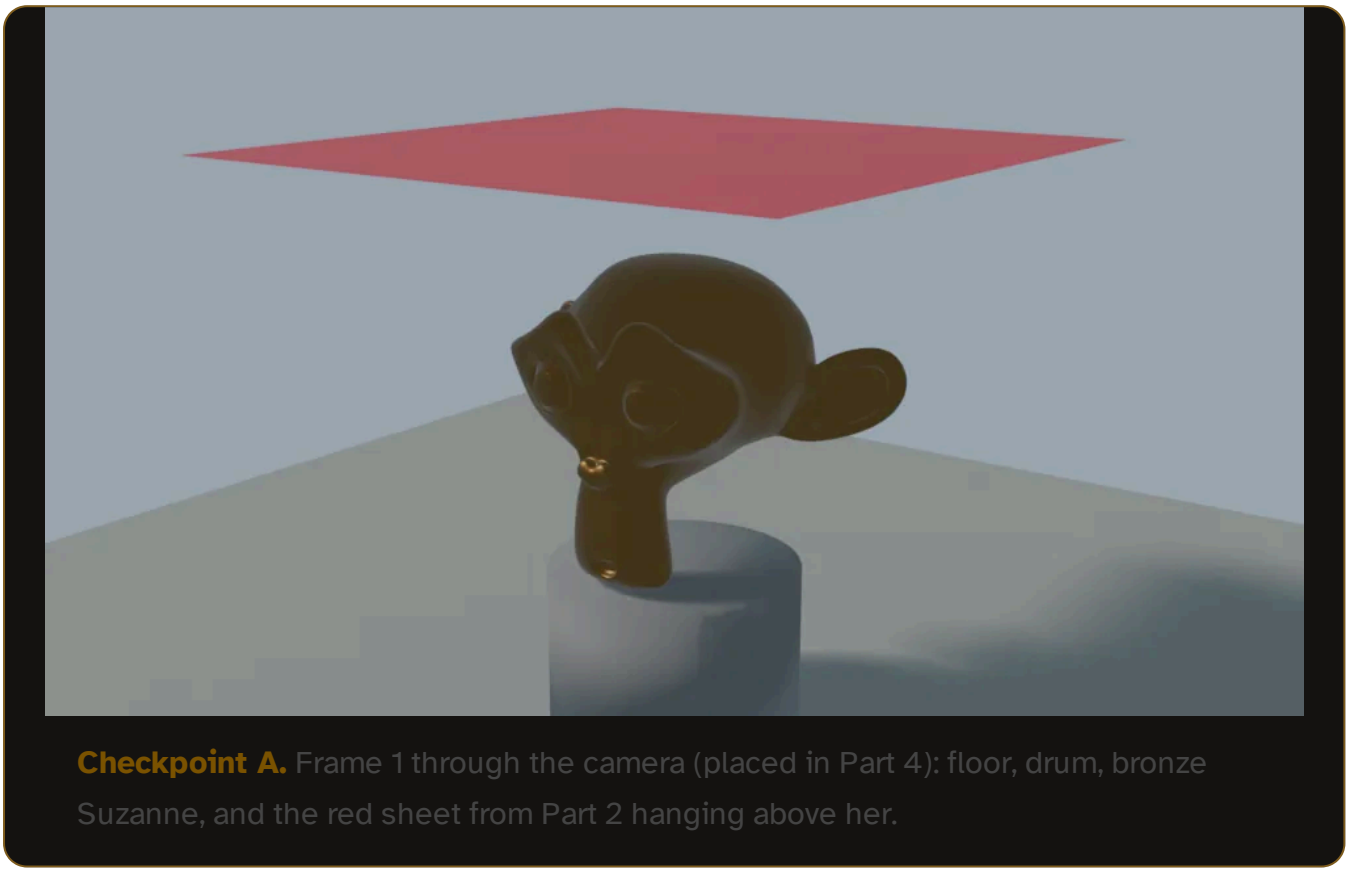
b Right-click › **Shade Smooth**. Wrench tab › **Add Modifier** › Generate › **Subdivision Surface**, **Levels Viewport** **2**.

c Material tab › New › Base Color Hex **7A5A2E**, Metallic **1**, Roughness **0.45**: bronze. F2, type **Suzanne**, Enter.

> **You should now see:** a smooth bronze monkey head on the drum, looking toward the front (-Y).

What does Subdivision Surface do?

> It splits every face into four and rounds the result, as many times as Levels says. Suzanne is a 500-face cartoon; two levels make her a smooth 8 000-face statue without you editing anything. The cloth gets the same modifier in Part 3, for the same reason.



Checkpoint A. Frame 1 through the camera (placed in Part 4): floor, drum, bronze Suzanne, and the red sheet from Part 2 hanging above her.

End of Part 1. Save.

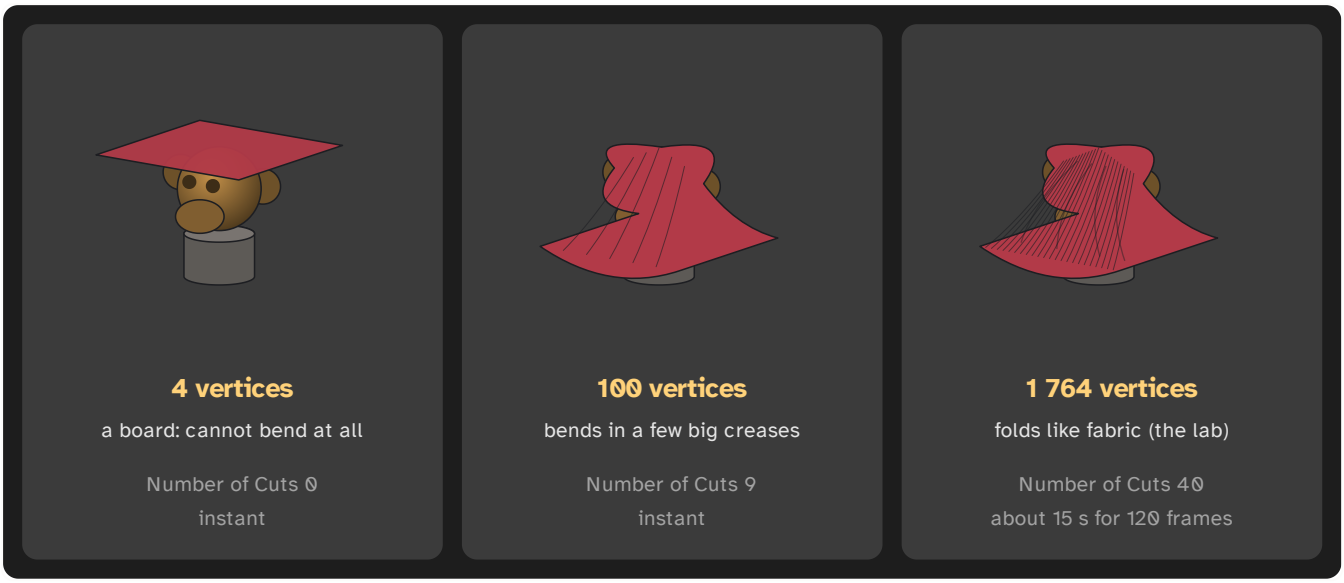
Part 2

A sheet that is cloth

Tools: **Subdivide** in Edit Mode, the **Physics** tab: **Cloth** and **Collision**, **Space**.

A plane becomes cloth in one click, but it can only bend where it has vertices: a four-vertex plane is a board. So the plane is cut into a 42 × 42 grid first. Then Cloth

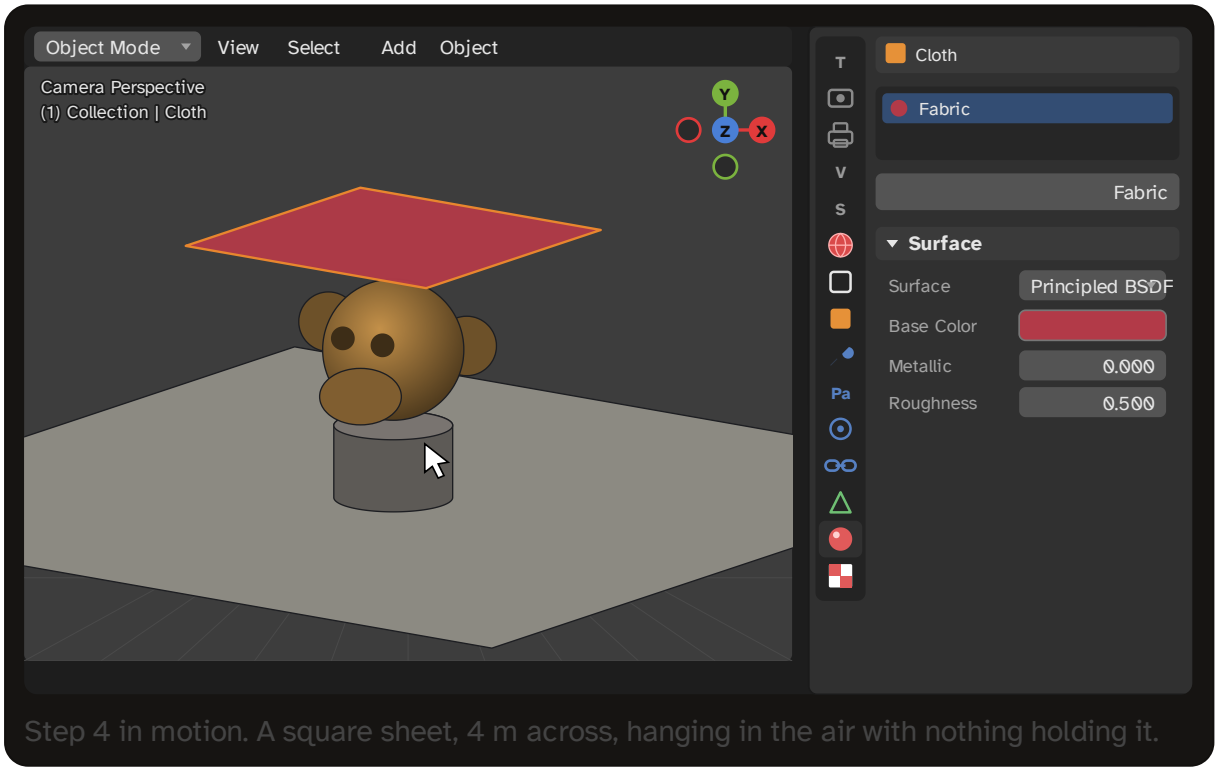
on the sheet, Collision on everything it should land on, and Space.



Cloth can only bend where it has vertices. More vertices, finer folds, slower simulation. The old sheet asked for three rounds of 10 cuts: 1.77 million vertices, which would never finish.

4 The sheet

The **Add** menu, the sidebar, the Material tab.



Step 4 in motion. A square sheet, 4 m across, hanging in the air with nothing holding it.

Do this:

a **Add** › **Mesh** › **Plane**. Sidebar: **Location** Z **3.6**. **Scale** X **2**, Y **2**, Z **2**: a 4 m square, 60 cm above the top of her head.

b F2, type **Cloth**, Enter. Material tab › New › Base Color Hex **B23A48**, Roughness **0.75**.

> **You should now see:** a flat red square floating above Suzanne, wider than she is on every side.

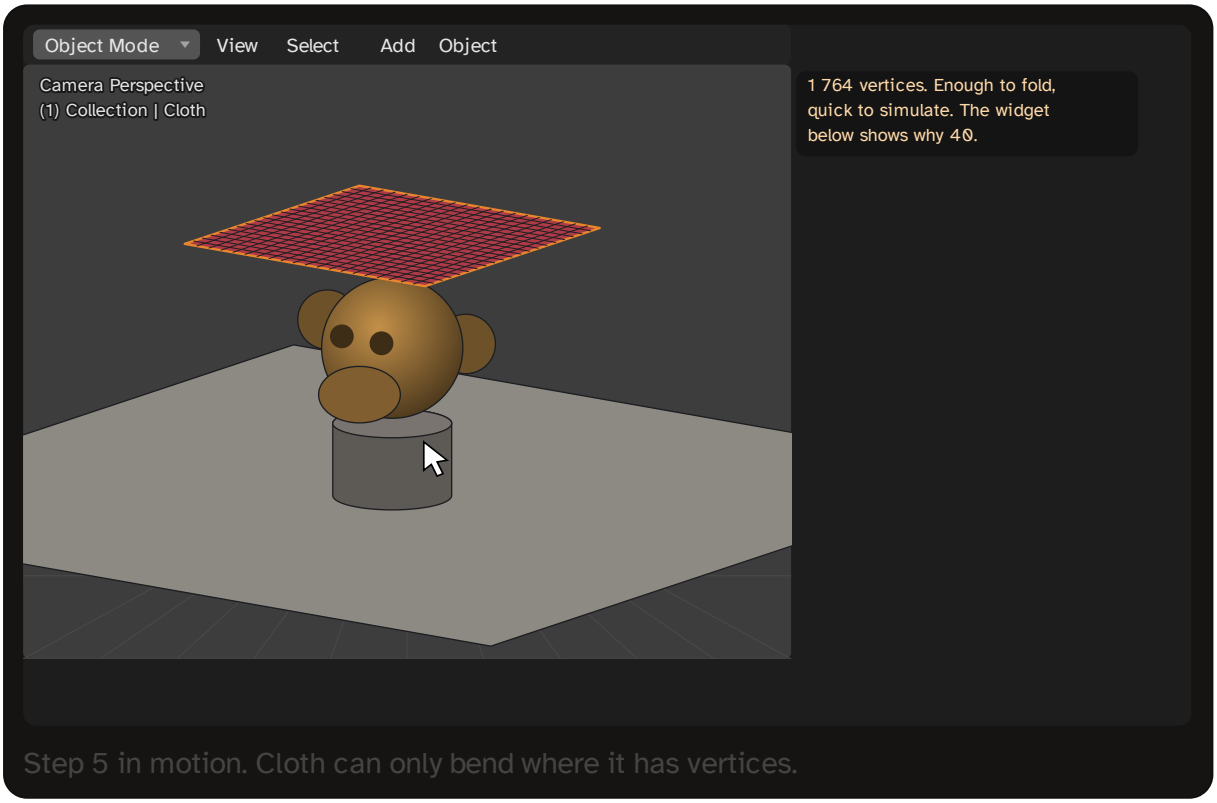
Why 4 m? Why 3.6 up?

> Suzanne is 2.7 m wide with her ears, so a 4 m sheet has 65 cm to hang down on each side. 3.6 is just above her head: far enough to fall visibly, close enough that it lands in the first second.

5 Cut it into a grid

Edit Mode, the right-click menu, the **Subdivide** panel bottom-left.

> **Predict first:** A plane has 4 vertices. After one Subdivide with 40 cuts, how many will it have? (Squares per side first, then vertices per side.)



Step 5 in motion. Cloth can only bend where it has vertices.

Do this:

a With Cloth selected, hover the viewport: Tab (Edit Mode). Everything is already selected.

b Right-click › **Subdivide**. Bottom-left, a small panel: **Number of Cuts** **40**. Do this before clicking anything else.

c Tab: back to Object Mode.

> **You should now see:** the square covered in a fine grid (in Edit Mode) and looking exactly as before in Object Mode.

The old sheet said 10 cuts, three times.

> Each round multiplies the squares per side: 11 × 11 × 11 = 1 331 per side, 1.77 million vertices. Blender would take hours per play and probably run out of memory. One round of 40 cuts gives 1 764 vertices: fine folds, 15 seconds per play. The widget below does the arithmetic.

How many cuts?

Number of Cuts sets how many vertices the sheet has, and how long each play takes.

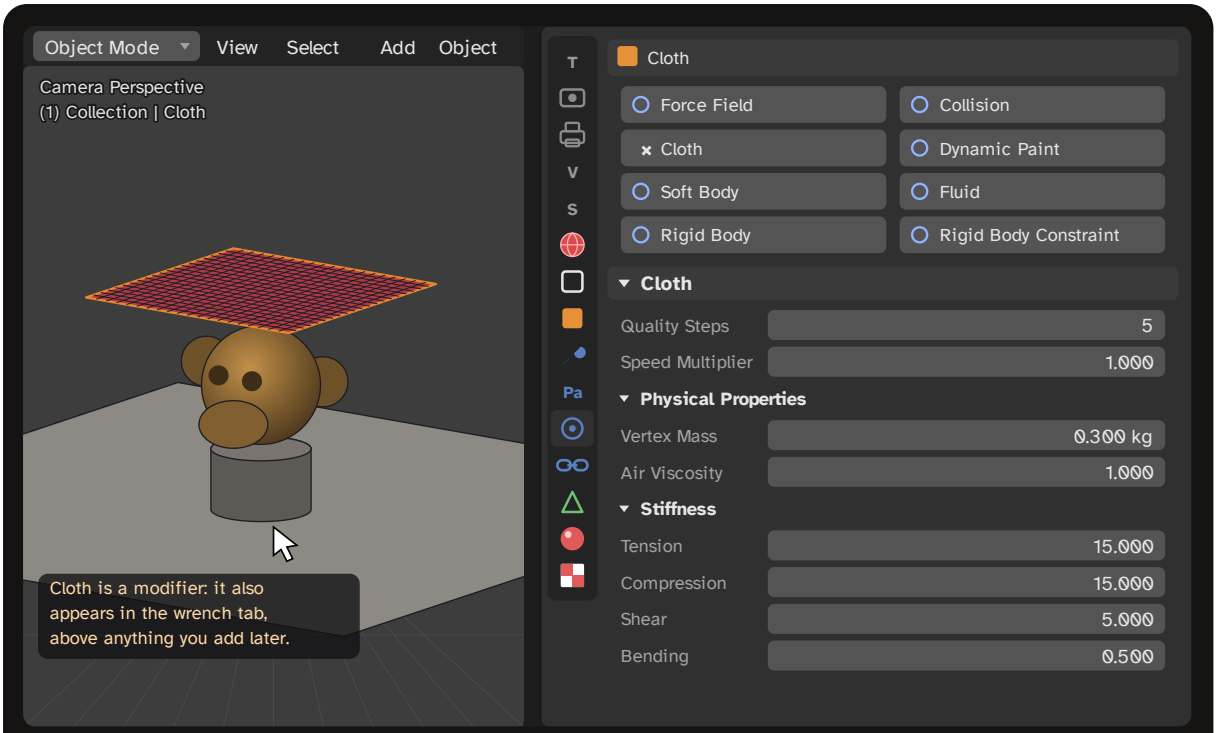


Number of Cuts **40** repeat three times (the old sheet)

Subdivide with **40 cuts**: 41 squares per side, **1,764 vertices**. About right: roughly 15 seconds to simulate 120 frames on a school machine.

6 Cloth

The **Physics** tab (blue orbit icon).



Step 6 in motion. One click makes the plane cloth; the numbers are cotton.

Do this:

a With Cloth selected: Physics tab › **Cloth**.

b Read the panel, change nothing: **Quality Steps** 5, **Vertex Mass** 0.3 kg, Stiffness **Tension** 15, **Bending** 0.5. That is cotton, the preset.

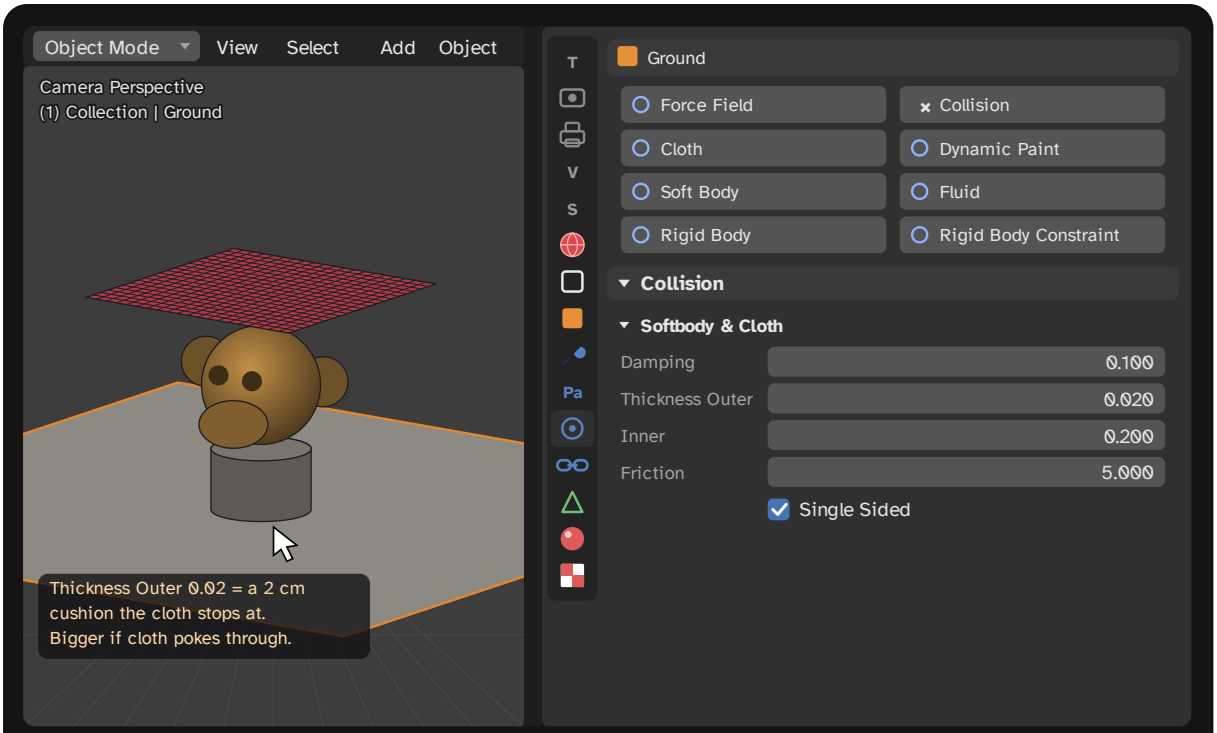
> **You should now see:** a long Cloth panel in the Physics tab, and no change in the viewport yet.

What is a preset?

> A set of numbers that imitates a real fabric. Cotton is the default. The dropdown at the top of the panel has Silk, Leather, Denim and Rubber too; try them in the extensions. The card above the steps says what each number means.

7 Collision on Suzanne, the drum and the floor

The Physics tab, three times.



Step 7 in motion. Collision on everything the cloth should land on.

Do this:

a Click **Suzanne**. Physics tab › **Collision**. A short panel: Thickness Outer 0.02. Leave it.

b Click the **Pedestal** › Physics › **Collision**. Click the **Ground** › Physics › **Collision**.

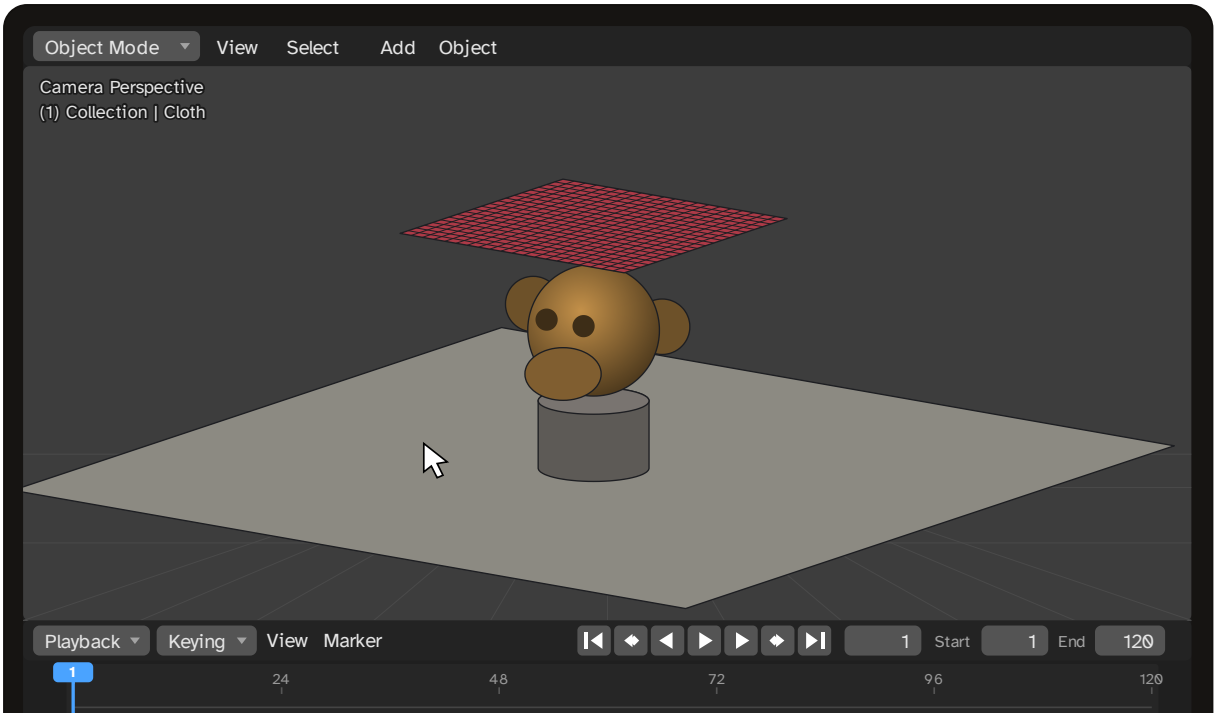
> **You should now see:** Collision listed in the Physics tab for all three. Nothing visible changes.

Why all three?

> Cloth only notices objects that have Collision; it falls straight through everything else. The old sheet only put Collision on Suzanne, so its cloth would have slid off her and fallen through the sphere and the floor. Any object can be a collider, and it costs nothing until cloth touches it.

8 Drop it

The Timeline, then **Space** over the viewport.



Step 8 in motion. Gravity, a sheet and a monkey.

Do this:

a Timeline: **End 120**.

b Hover the viewport, press **Space**. The sheet falls, hits her head, bends over her ears and slides down to the drum. By frame 60 it has stopped moving.

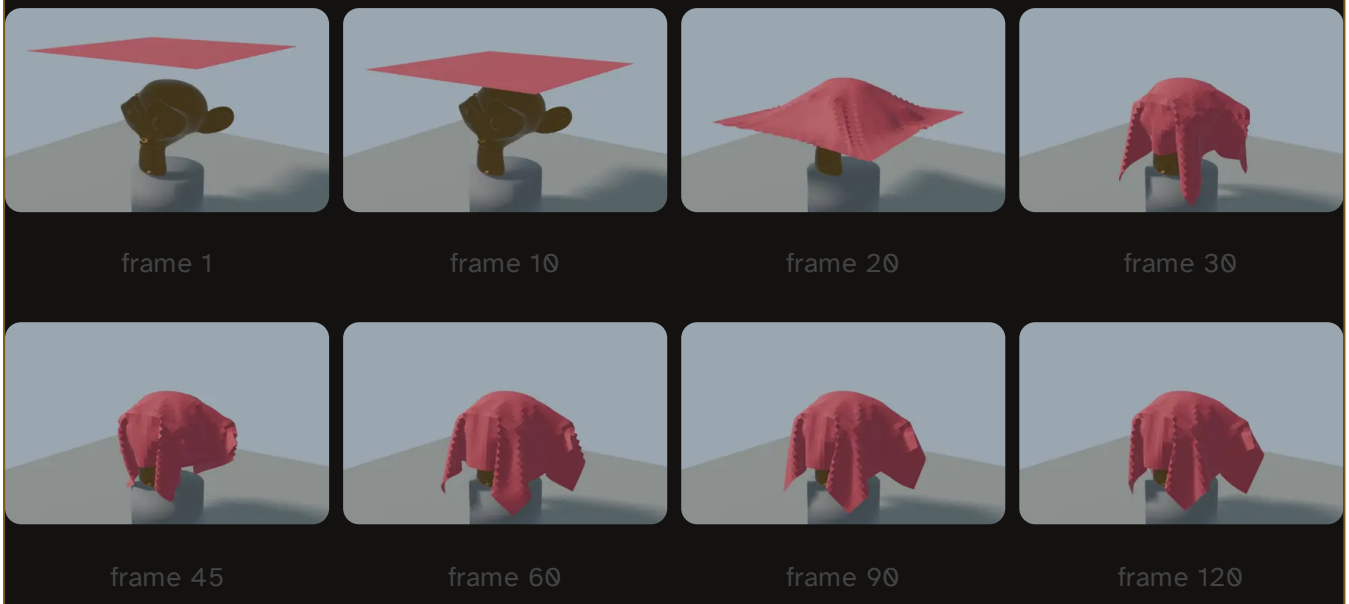
c **Space** pauses. **Shift** + ⌧ goes back to frame 1 and the sheet is flat again.

> **You should now see:** the drop and the drape, chunky and faceted for now. The first play is slow (Blender is simulating); the second is instant.

It went through her

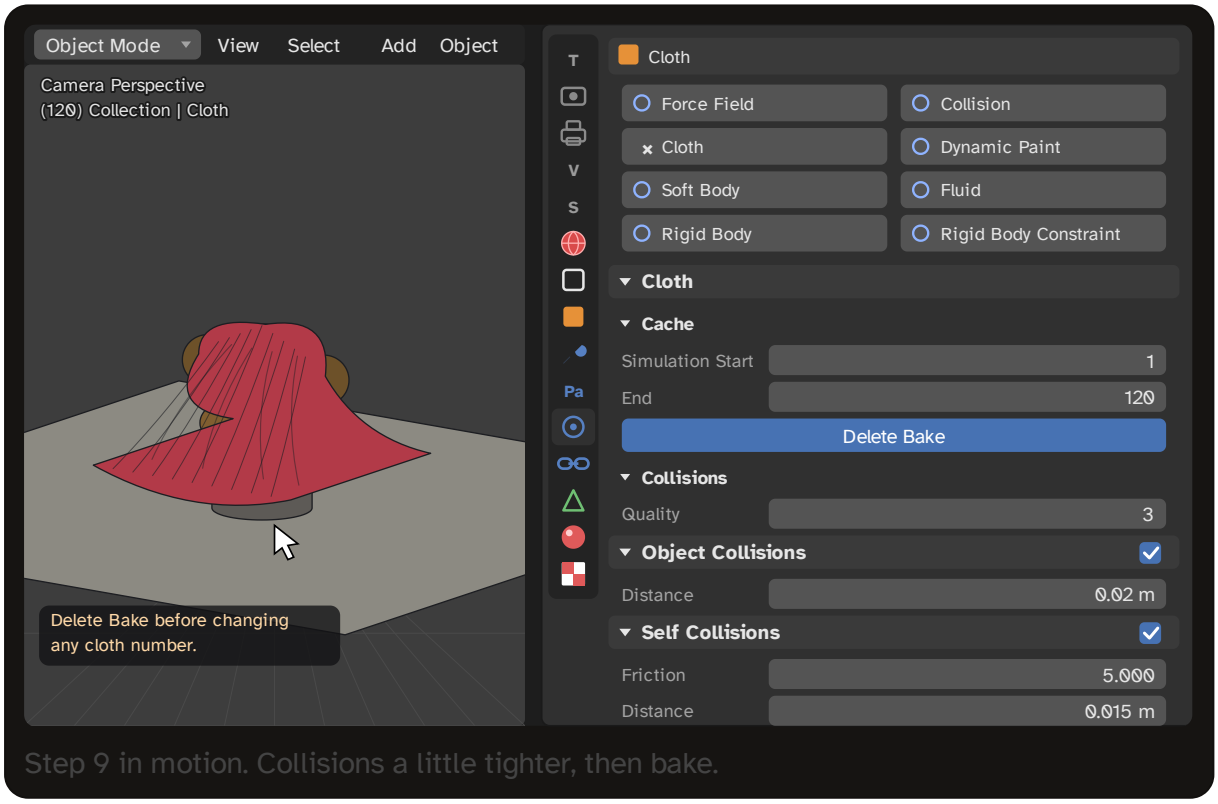
> Rarely at these settings, but if the sheet pokes through her ears or nose, Step 9 raises the collision quality and distance. If it fell straight through everything, a Collision is missing (Step 7).

Checkpoint B. Eight frames of the drop, rendered flat-shaded exactly as it looks in the viewport now.



9 Tighter collisions, then bake

The Physics tab › Cloth › **Collisions** and **Cache** panels (scroll down).



Step 9 in motion. Collisions a little tighter, then bake.

Do this:

a Collisions › **Quality 3**. Object Collisions › **Distance 0.02**. Tick **Self Collisions**.

b. Cache › **End 120**. Click **Bake**. A bar counts 120 frames.

c. Play again: identical every time, and scrubbing is instant.

> **You should now see:** the button reads Delete Bake, and the drape replays instantly.

What do those three do?

> Collision Quality is how many times per frame the cloth checks for things in its way; Distance is how far it keeps from them (2 cm). Self Collisions stops the sheet passing through its own folds where it bunches under her chin. Bake locks the result so the render matches, exactly as with the rigid bodies in Lab 1.

End of Part 2. Save.

Part 3

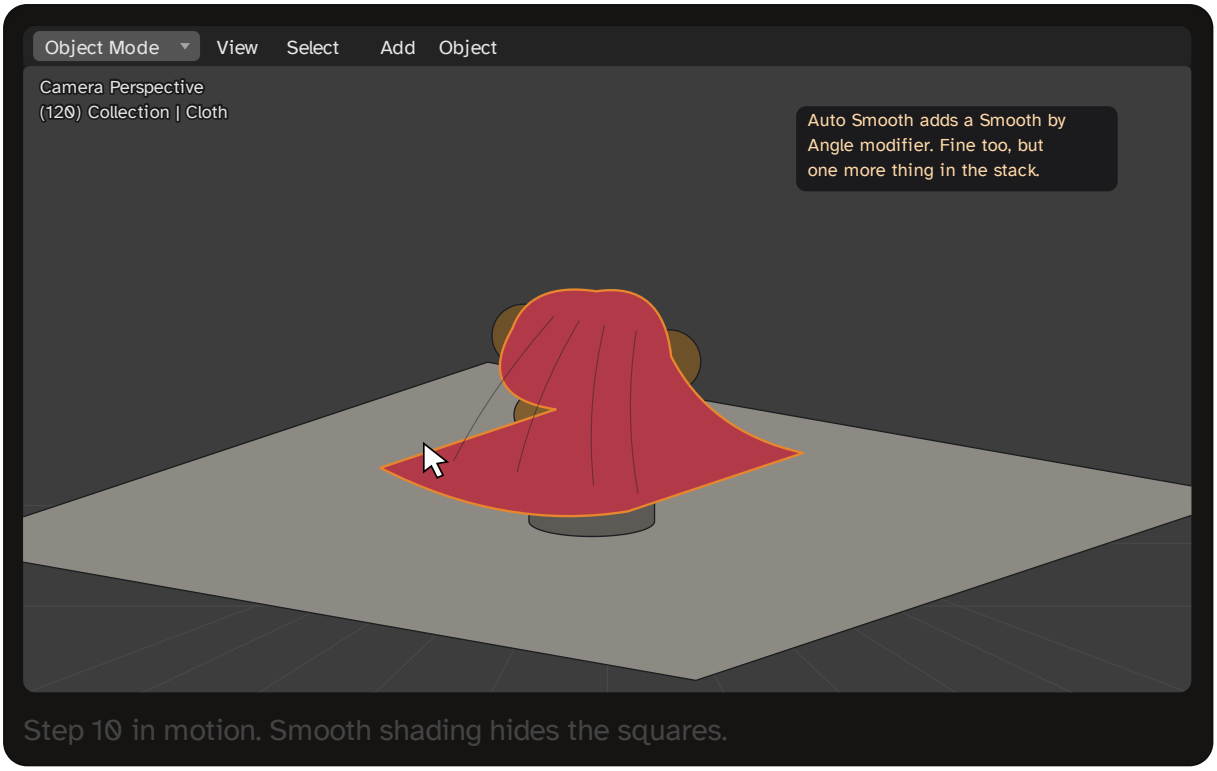
Smooth it

Tools: **Shade Smooth**, the **Subdivision Surface** modifier, the modifier order.

The simulation is right; it just looks like it was folded from paper. Two display tricks fix that without changing the simulation: smooth shading hides the facets, and a Subdivision Surface modifier rounds the folds.

10 Shade Smooth

The viewport at frame 120, the right-click menu.



Step 10 in motion. Smooth shading hides the squares.

Do this:

a. Go to frame **120** (type it in the frame box). The cloth is 1 681 flat squares and you can see every one.

b. Right-click the cloth › **Shade Smooth**.

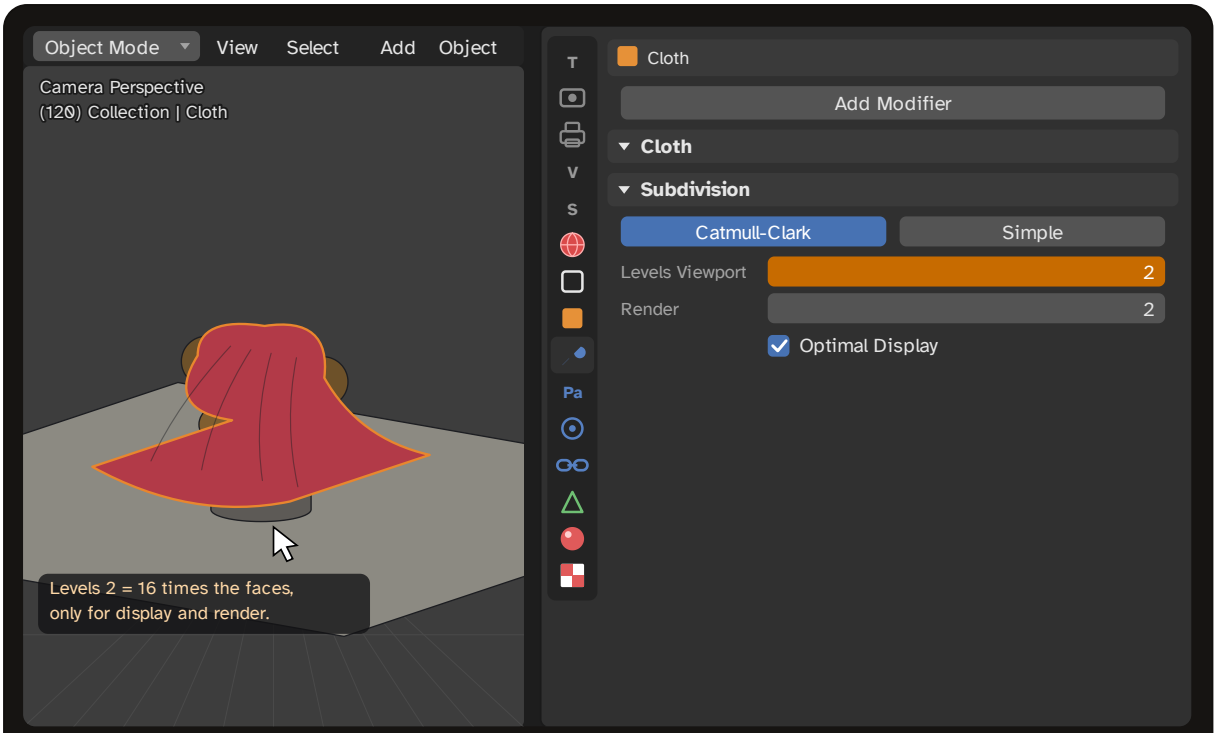
> **You should now see:** no visible facets; the folds read as one surface.

Shade Smooth or Shade Auto Smooth?

> Both hide the facets. Auto Smooth also keeps any edge sharper than 30° crisp by adding a Smooth by Angle modifier, which is what the old sheet used. Cloth has no sharp edges, so plain Shade Smooth does the job with nothing added to the modifier stack.

11 Subdivision Surface on the cloth

The **Modifiers** (wrench) tab.



Step 11 in motion. Simulate coarse, display fine.

Do this:

a. Wrench tab › **Add Modifier** › Generate › **Subdivision Surface**. **Levels Viewport** 2, **Render** 2.

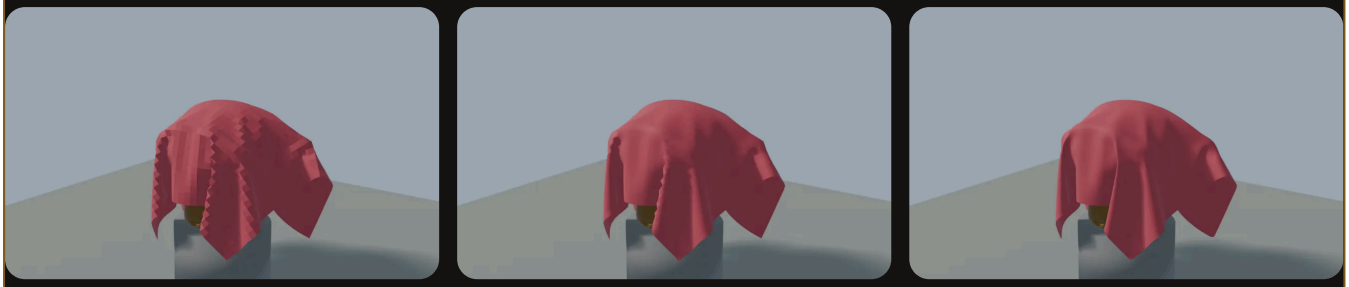
b. Check the order in the stack: **Cloth** on top, **Subdivision** below it. If not, drag the Subdivision header down (the dotted handle at its top-right).

> **You should now see:** rounder folds and a soft hem. Play it: the drape is the same, only smoother.

Why must Cloth be first?

> Modifiers run top to bottom. Cloth first means the simulation runs on the 1 764 real vertices and the subdivision smooths the result for display. The other way round, the cloth would simulate on 28 000 vertices: 16 times slower for no visible gain. Levels 2 is plenty; each level is four times the faces.

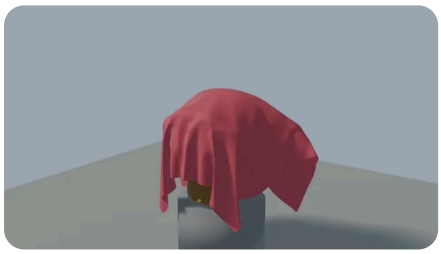
Checkpoint C. Frame 120 three ways: straight from the simulation, with Shade Smooth, and with Subdivision Surface level 2 as well. Same drape every time.



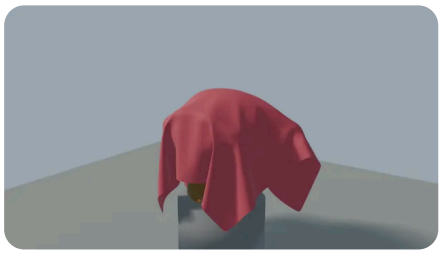
as simulated | + Shade Smooth | + Subdivision 2

What the cloth numbers do

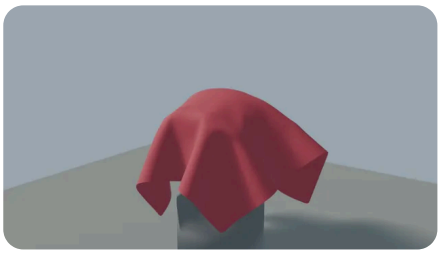
Rendered from the same scene with one number changed each time (Delete Bake, change, play, Bake). Try them yourself in the extensions.



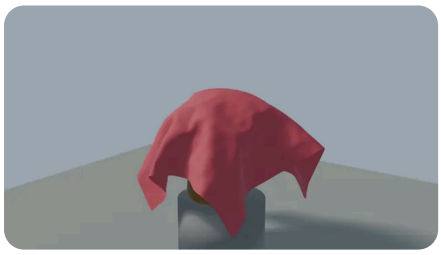
Bending 0.05: silk



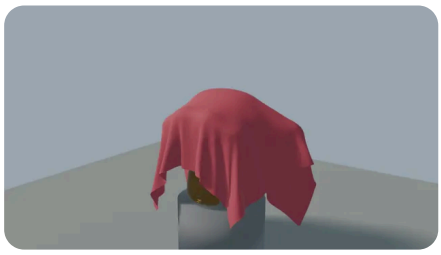
Bending 0.5: cotton (the lab)



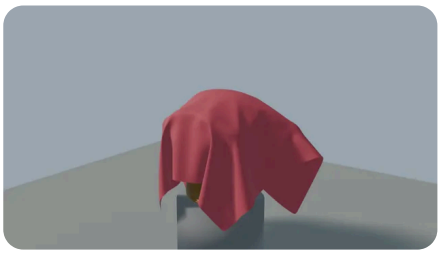
Bending 5: felt



Vertex Mass 0.05: tissue



Vertex Mass 3: wet canvas

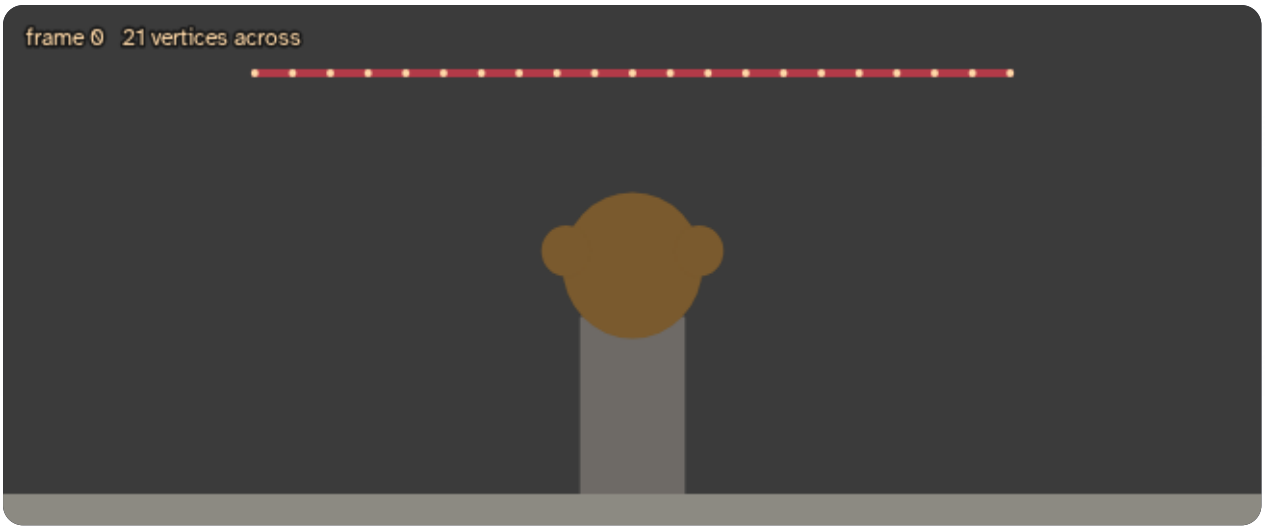


Self Collisions off: no visible change at this size

The last one is the honest surprise: with a cotton sheet this size, Self Collisions makes no difference you can see, because the folds never bunch deep enough to cross. It matters for silk, for a bigger sheet, or for a sheet landing on the floor in a heap, which is why Step 9 turns it on anyway.

Cloth lab

A strip of cloth over a head, side on. Change a number, drop it again.



Cuts 20 Bending 0.50 Vertex Mass 0.3 kg

Quality Steps 5

Drop Reset

20 cuts: the strip can only bend at its 21 points, fine enough to drape. **Bending 0.50**: cotton, it drapes with soft folds. **Quality 5**: enough checks per frame to stay calm. Side view only: Blender does this in 3D with 484 vertices for the same cuts.

End of Part 3. Save.

Part 4

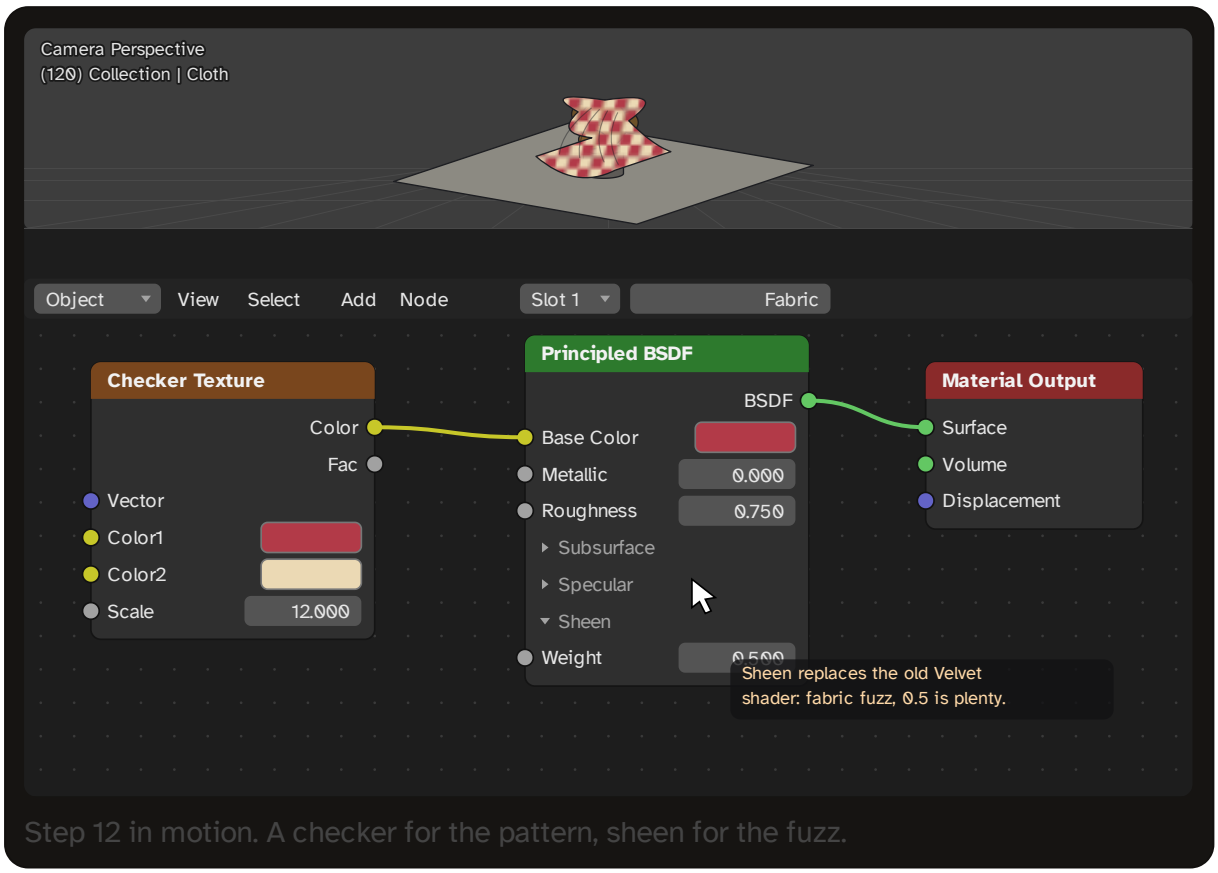
Fabric, light, camera, render

Tools: the **Shading** workspace, a **Checker Texture**, **Sheen**, the **World** and **Light** tabs, **F12**.

Plain red reads as plastic. A check pattern and a little sheen make it fabric. Then the same lighting rig as Lab 2, a camera with exact numbers, a still and a movie.

12 A check pattern with sheen

The **Shading** tab, the Shader Editor, the Principled BSDF node.



Step 12 in motion. A checker for the pattern, sheen for the fuzz.

Do this:

a. Click the cloth. **Shading** tab. Hover the Shader Editor: **Shift** + **A** › Texture › **Checker Texture**. Click it down left of the Principled BSDF.

b. In the node: **Scale** 12. **Color1** › Hex **B23A48**. **Color2** › Hex **EBD9B4**. Drag its **Color** dot to the Principled BSDF's **Base Color** dot.

c. In the Principled BSDF, click **Sheen** to open it › **Weight** 0.5.

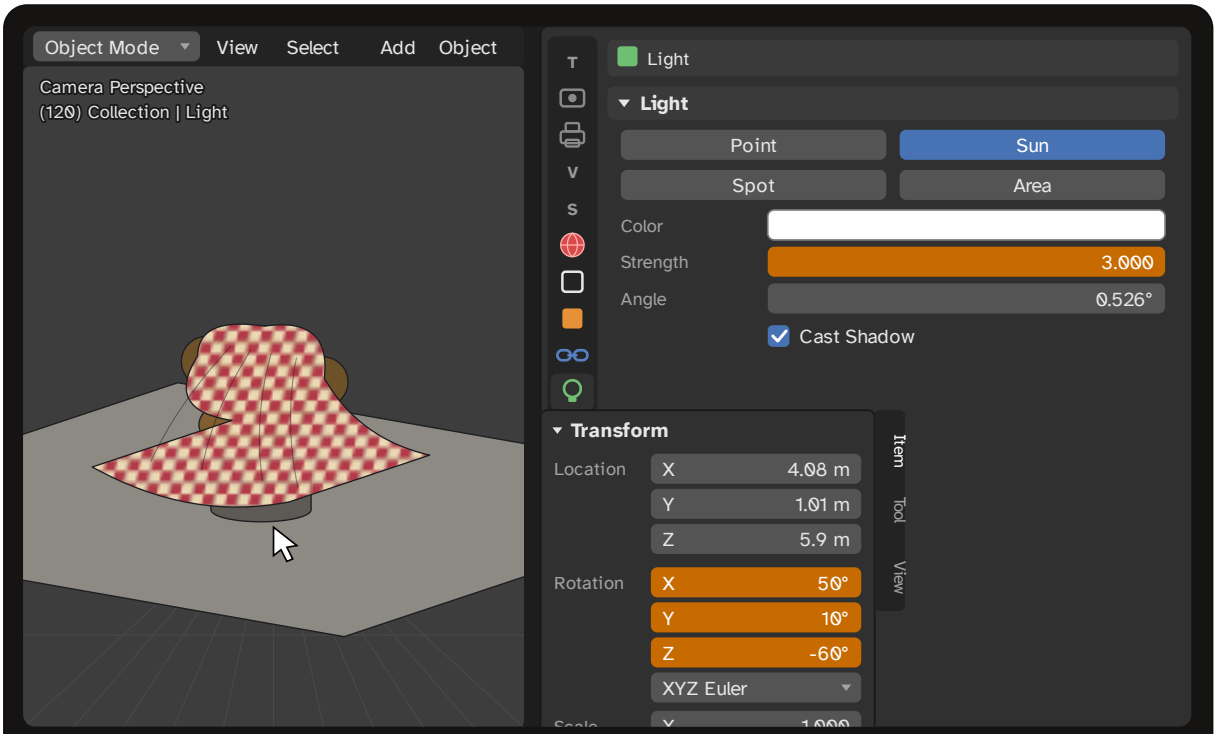
> **You should now see:** a red-and-cream check that follows every fold, with a soft pale glow along the edges of the folds in Material Preview.

What is Sheen?

> The faint fuzz-glow at the edges of fabric, where tiny fibres catch the light. The old sheet suggested the Velvet BSDF for this; that shader was removed in Blender 4.0 and Sheen on the Principled BSDF is its replacement. 0.5 is a cotton amount; 1 is velvet.

13 World and sun

The **World** tab (red globe), the Outliner, the green light **Data** tab, the sidebar.



Step 13 in motion. A pale sky and a sun.

Do this:

- **a** World tab › Surface › **Color** › Hex **9FB4C8**. Strength **1**.
- **b** Outliner › **Light**. Green bulb tab › **Sun**. **Strength 3**.
- **c** Sidebar › **Rotation** X **50**, Y **10**, Z **-60**.

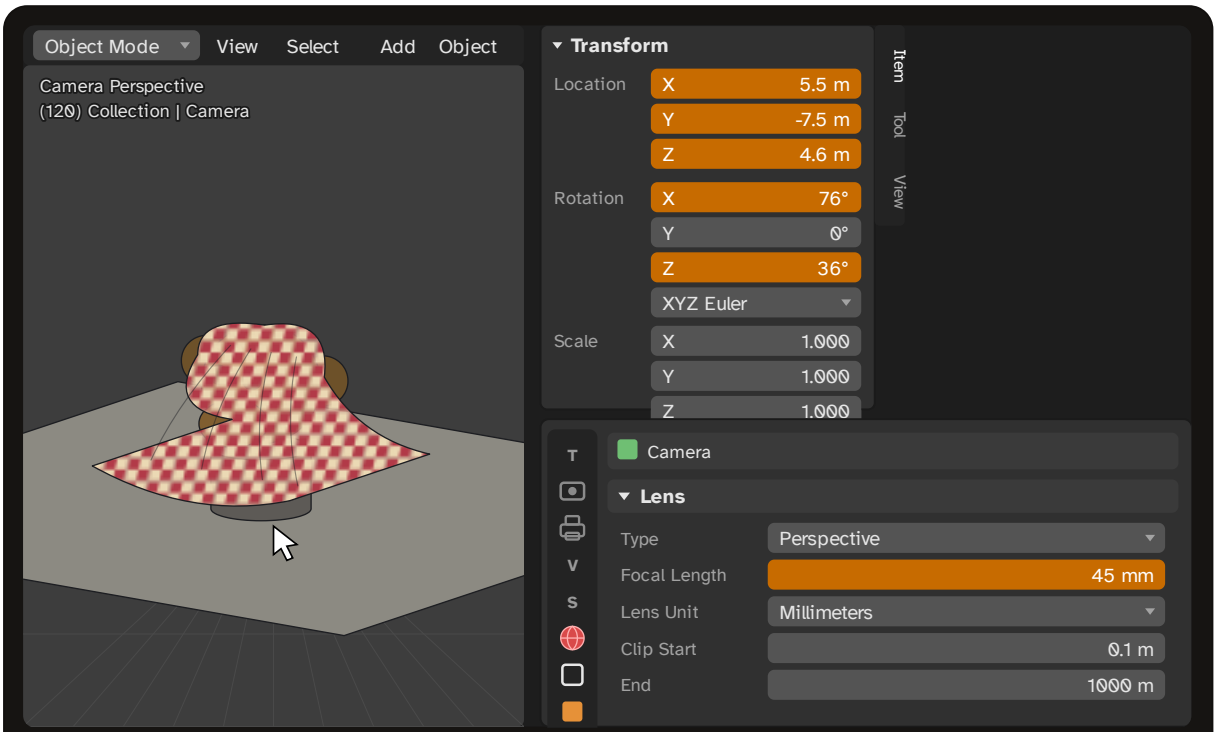
> **You should now see:** in Rendered shading (fourth sphere): a pale sky, a sun from the front-left, crisp fold shadows.

Same numbers as Lab 2?

> Yes, on purpose: a pale world plus a 3-strength sun from the front-left is a dependable daylight rig. Reuse it whenever a render comes out dark.

14 Place the camera

The Outliner, the sidebar, the green camera **Data** tab, **Numpad 0**.



Step 14 in motion. Three numbers for where, three for which way, one for the lens.

Do this:

a. Outliner › **Camera**. Sidebar **Location** X **5.5**, Y **-7.5**, Z **4.6**. **Rotation** X **76**, Y **0**, Z **36**.

b. Camera tab › **Focal Length** **45**. **Numpad 0** to look through it.

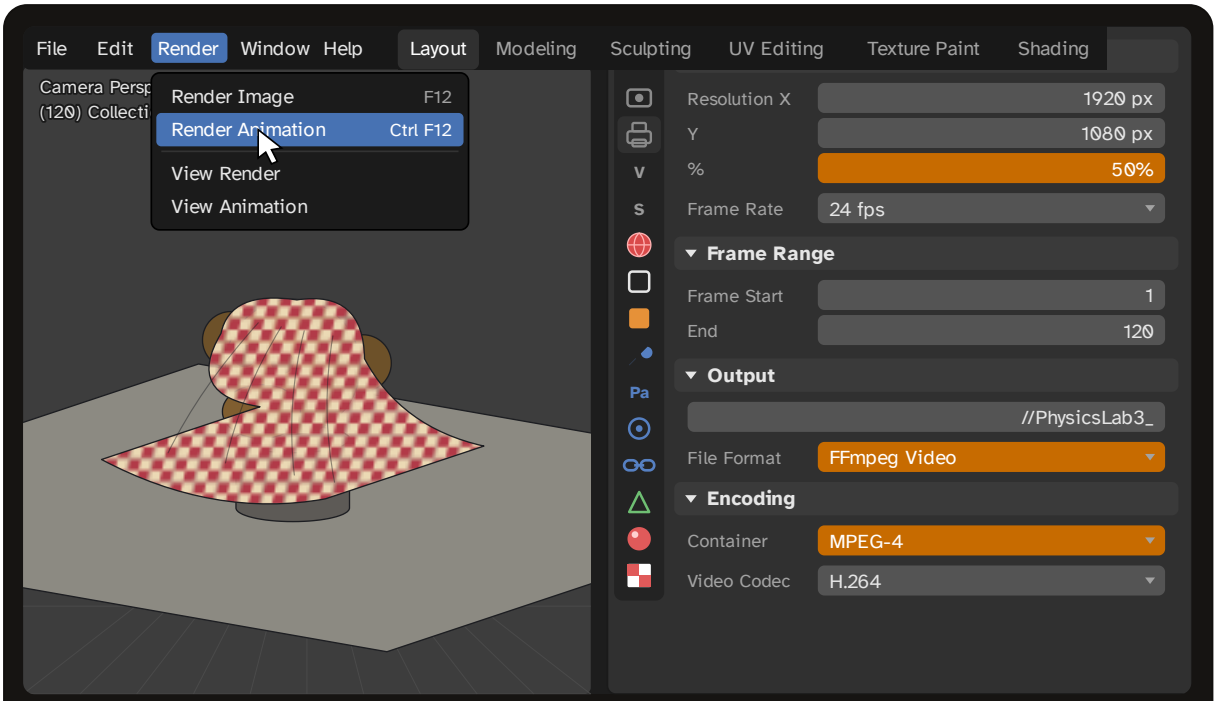
> **You should now see:** the drum, Suzanne and the whole sheet in frame, the hem clear of the bottom edge.

Why those numbers?

> The camera stands 9 m away, front-right, 4.6 m up, turned 36° to look at the drum and tilted down 14°. 45 mm is a little wider than the default 50 so the 4 m sheet fits at frame 1.

15 Render: a still and the drop

The frame box, **F12**, the render window's **Image** menu, the **Output** tab, the **Render** menu.



Step 15 in motion. A still for the wall, a movie for the drop.

Do this:

a. Frame **120**. Hover the viewport, **F12**. In the render window: **Image** › **Save As** › PNG, name it **PhysicsLab3_DrapedCloth_YourName_render**. **Esc**.

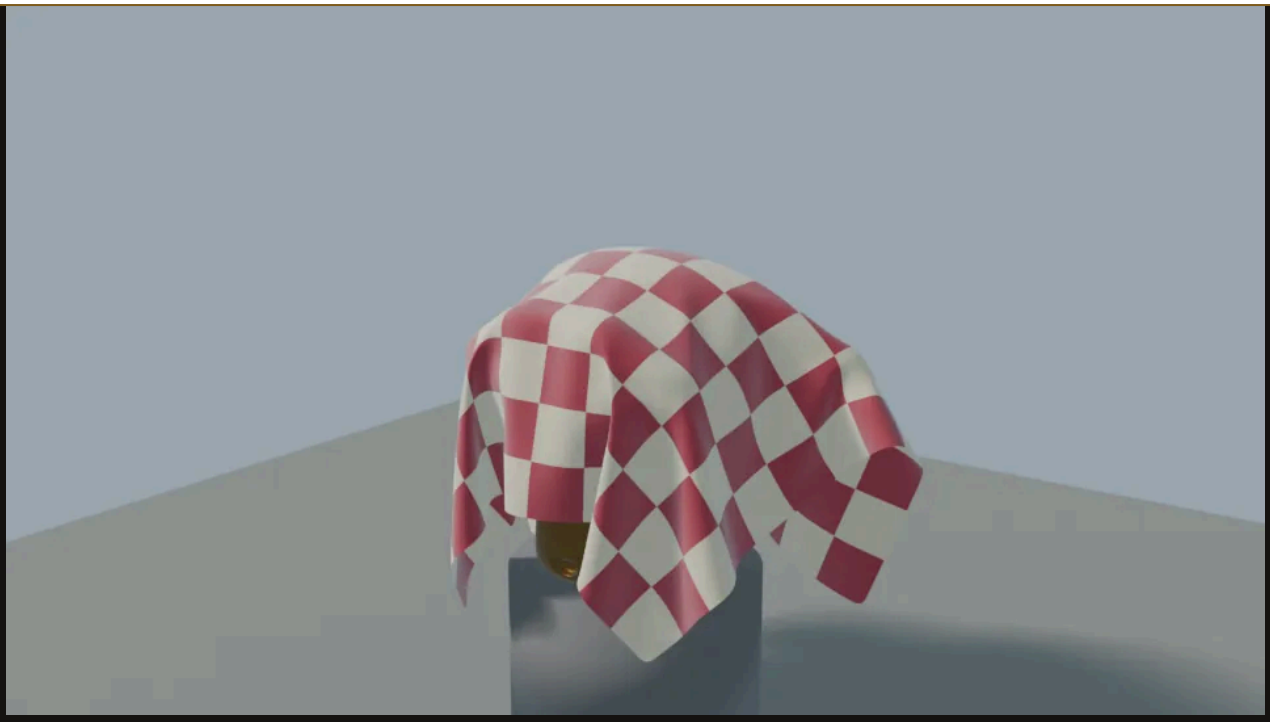
b. Output tab: **Resolution % 50**, **File Format** FFmpeg Video, **Container** MPEG-4, output folder your Physics labs folder. **Render › Render Animation** (or **Ctrl**+**F12**). A few minutes.

c. **Ctrl**+**S**. Upload the PNG, the .mp4 and the .blend to SEQTA.

> **You should now see:** a PNG matching Checkpoint D and a 5-second movie of the sheet dropping.

Why both?

> The still is for looking closely at folds and fabric; the movie is the physics. A cloth that lands well is only impressive if you see it land.



Checkpoint D. Frame 120 through the camera: checked cotton draped over bronze Suzanne, lit by the sky and the sun.



Checkpoint D (close). Close up: the check follows the folds, and the sheen catches the light along their edges.

End of the lab.

Part 5

Finished early? Pick one

Tools: everything above, used again.

Rule for every experiment: **Delete Bake**, change one thing, play from frame 1, **Bake** again if you like it.

- **Try the presets**

  Top of the Cloth panel, the preset dropdown: **Silk**, **Leather**, **Denim**, **Rubber**. Predict first how each will drape, then play. Silk needs Self Collisions on; Rubber may need Quality Steps 10. Which one looks most like a bedsheet? Which like a tarpaulin?

- **Pin a corner**

  Tab into Edit Mode on the cloth, click one corner vertex, **Object Data tab (green triangle) › Vertex Groups › +**, name it **Pin**, click **Assign**. Tab out. Cloth panel › Shape › **Pin Group**: Pin. Play: the sheet hangs from that corner like a flag on a hook instead of falling. Pin two corners for a washing line.

- **Wind**

  Add › Force Field › Wind, Location **-5, 0, 2**, Rotation Y **90** (so it blows along +X), Physics tab › Strength **800**. With a pinned corner the sheet streams out like a flag. Try Strength 200 for a breeze and add Turbulence (Force Field type) for flapping.

- **A thicker sheet**

  Real cloth has thickness; a plane has none, so the hem looks paper-thin close up. Wrench tab › Add Modifier › Generate › **Solidify**, Thickness **0.01**. Put it **after** Cloth and **before** Subdivision in the stack (drag the headers). The edges of the sheet gain a visible edge.

- **Something else under the sheet**

  Hide Suzanne (the eye icon in the Outliner), add a UV Sphere and a Cube as new colliders at different heights, Delete Bake, play. A sheet over a cube shows every edge; over a sphere it slides off unless Friction (Collision panel on the sphere) is raised to 20.

Something’s not right

Find the problem that matches yours.

- **I can’t orbit (no middle mouse button)**

  Drag on the coloured gizmo itself to orbit. Or turn on **Edit** › **Preferences** › **Input** › **Emulate 3 Button Mouse**, then Alt + left-drag orbits.

- **The cloth falls straight through everything**

  The objects under it have no Collision. Click each one (Suzanne, Pedestal, Ground), Physics tab › Collision. Then Shift+Left and play again.

- **The cloth pokes through her ears or nose**

  Step 9: Collisions › Quality 3 and Distance 0.02, then Delete Bake and play. Still poking? Quality 4, or on Suzanne's Collision panel raise Thickness Outer to 0.05.

- **The folds pass through each other under her chin**

  Self Collisions is off. Tick it in the Cloth panel › Collisions, Delete Bake, play.

- **The sheet lands like a stiff board**

  It was not subdivided (Step 5), or the subdivide happened on another object. Click the cloth, Tab: you should see a fine grid. If not, Subdivide again with 40 cuts.

- **Playing takes forever**

  Too many vertices: Subdivide was run more than once, or with a huge number of cuts. Click the cloth, Tab, select all (A), X › Vertices; add a new plane and redo Steps 4–6. The widget in Part 2 shows how fast the count grows.

- **Nothing moves when I press Space**

  The mouse must be over the viewport (or the Timeline) when you press Space. Also check the End frame is 120, not 1.

- **Everything jumps back when I click**

  That is frame 1. The cloth is only draped at later frames: drag the Timeline marker or play.

- **Bake is greyed out**

  The file is not saved. Ctrl+S, then Bake.

- **I changed a number and nothing happened**

  The simulation is baked. Delete Bake, play from frame 1 (the change shows), then Bake again.

- **Subdivision made playback very slow**

  The Subdivision modifier is above Cloth in the stack, so the cloth is simulating 28 000 vertices. Wrench tab: drag Subdivision below Cloth (the dotted handle at the top-right of its header).

- **The cloth still looks faceted after Shade Smooth**

  Shade Smooth was applied to another object (Suzanne, or the floor). Click the cloth in the viewport first, then right-click › Shade Smooth.

- **The cloth shivers or jitters after it lands**

  Quality Steps too low for the collisions. Cloth panel › Quality Steps 8, Delete Bake, play. Raise Air Viscosity to 2 if it still rocks.

- **The check pattern is stretched or missing**

  The Checker Texture Color dot must go to Base Color, not to Metallic or Roughness. Stretched squares mean the Scale is too low or the wire went to the wrong socket: drag it off and redo.

- **The render is dark**

  Step 13: world colour 9FB4C8 at Strength 1, and the Light set to Sun, Strength 3, Rotation 50, 10, -60.

- **The render shows the flat sheet**

  The frame box was on 1. Set it to 120 before F12.

- **My render disappeared**

  A render is only kept if you use Image › Save As in the render window before closing it.

Eight quick questions

Pick an answer to lock it in. You'll find out straight away whether you're right.

1 Why subdivide the plane before adding Cloth?

- Cloth needs at least 1 000 vertices
- Cloth can only bend where it has vertices
- Subdivide adds the Cloth physics
- It makes the plane bigger

2 The old sheet said Subdivide with 10 cuts three times. How many vertices is that, roughly?

- About 1 000
- About 30 000
- About 1.8 million
- About 121

3 What does Collision on Suzanne do?

- Makes her fall
- Tells cloth (and other soft things) to stop at her surface
- Makes her a rigid body
- Smooths her

4 Which number makes cloth behave like silk instead of cotton?

- Bending 0.05
- Quality Steps 1
- Vertex Mass 3
- Scale 12

5 Self Collisions stops

- the cloth hitting Suzanne
- the cloth passing through its own folds
- the cloth falling
- the bake

6 Shade Smooth changes

- the mesh
- the simulation
- only how light is shaded across the faces
- the number of vertices

7 Why must the Cloth modifier sit above Subdivision Surface?

- Blender insists
- So the simulation runs on the coarse mesh and the subdivision smooths the result
- So the cloth is heavier
- It does not matter

8 What replaced the Velvet BSDF for fabric fuzz?

- Glossy BSDF
- Sheen on the Principled BSDF
- A Noise Texture
- Metallic 1

Check yourself

Tick each one you can honestly do now. Anything left unticked is what to ask about next lesson.

- [ ] I can build a simple scene with exact numbers and smooth it with Shade Smooth and Subdivision Surface.
- [ ] I can subdivide a plane in Edit Mode and explain why vertex count matters for cloth.

- [ ] I can add Cloth and Collision and say what Bending, Vertex Mass, Quality and Self Collisions do.
- [ ] I can bake a cloth simulation and know when to delete the bake.
- [ ] I can put a modifier in the right order in the stack and explain why.
- [ ] I can give cloth a texture and sheen, light the scene, and render a still and a movie.

Think about it

A real bedsheet is about 1.5 mm thick and the cloth here is a surface with no thickness at all. Where in the result can you see that it has no thickness, and which extension would fix it?

Upload to SEQTA — Physics Lab 3

Upload **PhysicsLab3_DrapedCloth_YourName.blend**, **PhysicsLab3_DrapedCloth_YourName_render.png** and the **.mp4**. The .blend should contain the subdivided Cloth with a baked cache, Collision on Suzanne, the pedestal and the ground, the Subdivision modifier below Cloth, the checker material, the world colour, the sun and the camera.

Due one week from when this lab was handed out.

– Teacher notes

Timing

Two lessons. Lesson 1: Parts 1–2 (scene, cloth, drop, bake). Lesson 2: Parts 3–4 (smoothing, material, light, render) and extensions. A 120-frame play with 1 764 vertices takes about 15 s on a school machine; the 50% EEVEE movie about 4 minutes.

What changed from the original sheet

- **Subdivision count.** The original asked for Number of Cuts 10 “and repeat two more times”, which is 1.77 million vertices and will not simulate on any school computer. One Subdivide with 40 cuts (1 764 vertices) is used, with a widget and a card so students see why the count matters both ways.
- **Colliders.** Only Suzanne had Collision in the original; the sheet would slide off her and fall through the sphere and the floor. The sphere “as a floor/base” is replaced by a floor plane and a cylinder pedestal, all three with Collision.
- **Exact numbers everywhere** (sheet 4 m at Z 3.6, Suzanne at Z 2 on a 1 m drum, camera, lights) replace “press S then drag” and “position Suzanne sitting on the sphere”. All were built from these steps in Blender 5.2.2 (bpy) and simulated; the checkpoint pictures and comparison renders are EEVEE renders of that file.
- **Collision tuning is explicit** (Quality 3, Distance 0.02, Self Collisions on) rather than “increase the Collision Distance in her settings”, and the bake is taught, as in Lab 1, so renders match playback.
- **Modifier order** (Cloth above Subdivision) is stated and explained; the original did not mention it and a reversed stack is the most common cause of a crawling viewport.
- **Shade Smooth** is used instead of Shade Auto Smooth (both work; Auto Smooth adds a Smooth by Angle modifier the cloth does not need). The challenge “try the Velvet BSDF” is replaced by Principled Sheen,

since Velvet was removed in Blender 4.0. Wave Texture is replaced by a Checker Texture, which shows the folds more clearly.

Where students usually get stuck

Subdividing with the whole mesh not selected (only part of the plane gets cut). Forgetting Collision on the floor. Pressing Space with the mouse over the Properties editor. Changing a cloth number while baked and seeing nothing happen. Adding Subdivision before Cloth. Rendering on frame 1.

Quiz answers

1 Only bends where vertices are. 2 About 1.8 million. 3 Cloth stops at her surface. 4 Bending 0.05. 5 Passing through its own folds. 6 Only the shading. 7 Simulate coarse, smooth after. 8 Sheen.

Quick rubric

Criterion	Developing	Achieved	Extending
Setup	Plane dropped, falls through objects	Subdivided sheet, Cloth, Collision on all three objects, drapes and settles	A preset, pin group or wind used with a predicted and observed result
Quality	Faceted, pokes through, unbaked	Collisions tuned, Self Collisions, baked; Shade Smooth and Subdivision in the right order	Solidify for thickness, or a different collider with friction tuned
Output	Screenshot only	Checker material with sheen, lit, PNG and .mp4 submitted	Material or lighting choices explained in the submission

Every picture on this page was rendered in Blender from the steps you just followed.